I2C Communication algorithms

For RTC(Digital Calender) Slave
DS1307

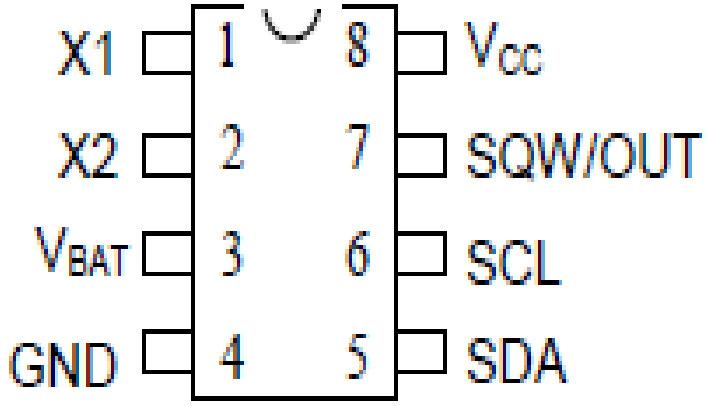
(Referring Data Sheet for DS1307)

FEATURES

- Real-time clock (RTC) counts seconds, minutes, hours, date of the month, month, day of the week, and year with leap-year compensation valid up to 2100
- 56-byte, battery-backed, nonvolatile (NV) RAM for data storage
- Two-wire serial interface
- Programmable squarewave output signal

The DS1307 operates in the regular mode (100kHz) only.

Pin diagram

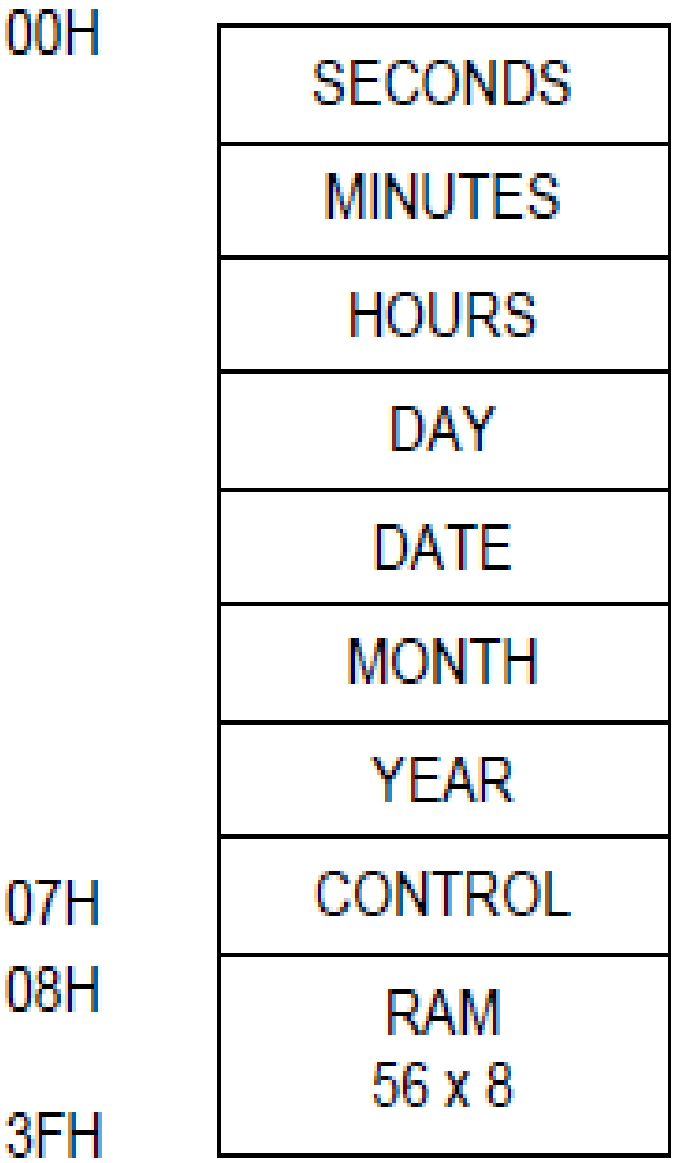


DS1307 8-Pin DIP (300-mil)

PIN DESCRIPTION

V_{CC}	- Primary Power Supply
X1, X2	- 32.768kHz Crystal Connection
V_{BAT}	- +3V Battery Input
GND	- Ground
SDA	- Serial Data
SCL	- Serial Clock
SQW/OUT	- Square Wave/Output Driver

DS1307 ADDRESS MAP



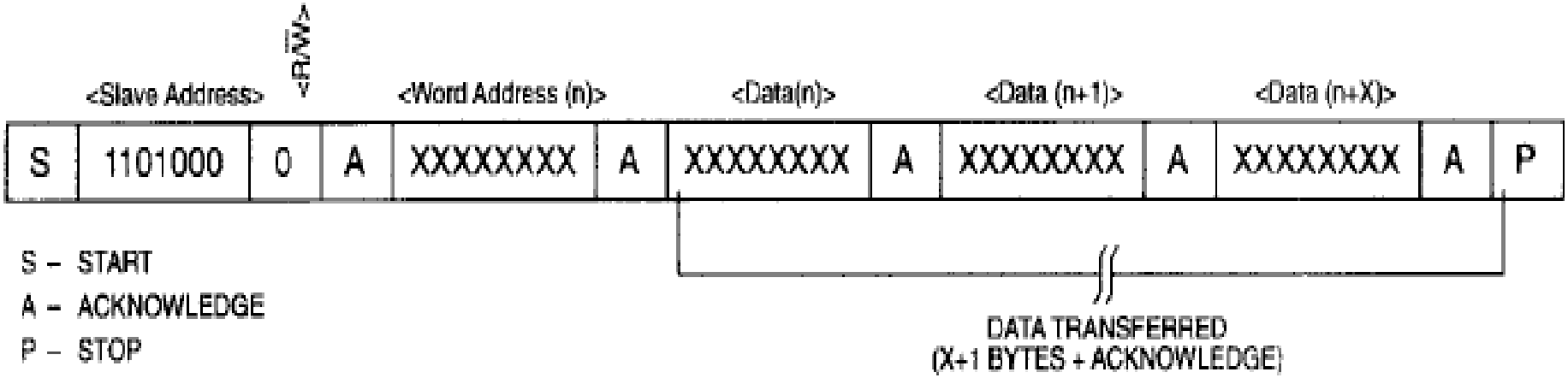
DS1307 TIMEKEEPER REGISTERS

	BIT7							BIT0	
00H	CH	10 SECONDS			SECONDS				00–59
	0	10 MINUTES			MINUTES				00–59
	0	12/24	10 HR A/P	10 HR	HOURS				01–12 00–23
	0	0	0	0	0	DAY			1–7
	0	0	10 DATE		DATE				01–28/29 01–30 01–31
	0	0	0	10 MONTH	MONTH				01–12
	10 YEAR				YEAR				00–99
07H	OUT	0	0	SQWE	0	0	RS1	RS0	

SQUAREWAVE OUTPUT FREQUENCY

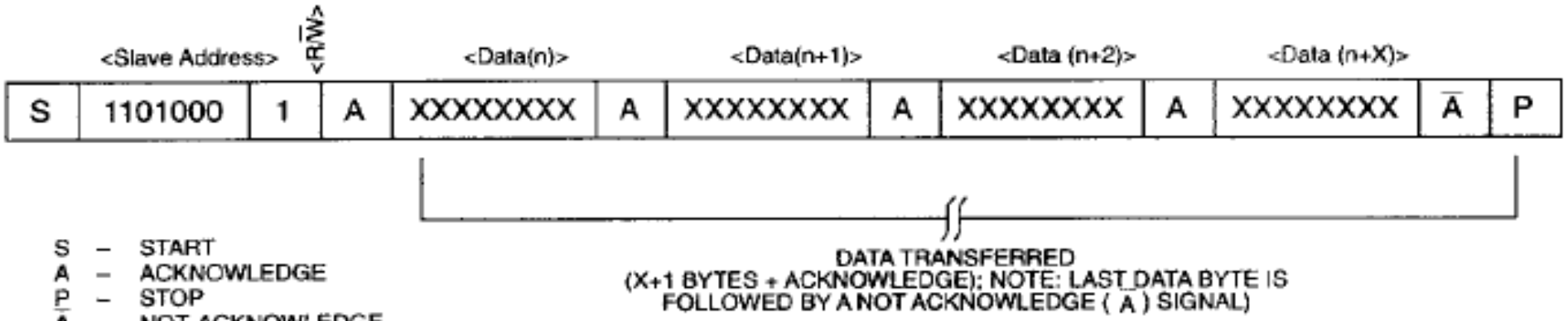
RS1	RS0	SQW OUTPUT FREQUENCY
0	0	1Hz
0	1	4.096kHz
1	0	8.192kHz
1	1	32.768kHz

I2C Communication Algorithms



S – START
A – ACKNOWLEDGE
P – STOP

*R/$\overline{W}$ – READ/WRITE OR DIRECTION BIT. ADDRESS = D0h



S – START
A – ACKNOWLEDGE
P – STOP
Ā – NOT ACKNOWLEDGE

*R/$\overline{W}$ – READ/WRITE OR DIRECTION BIT ADDRESS = D1h

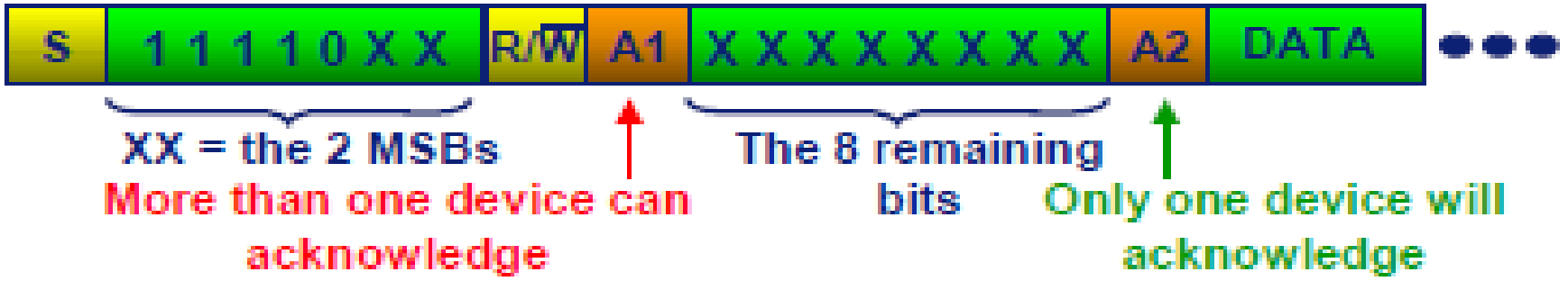
I²C Address, 7-bit and 10-bit formats

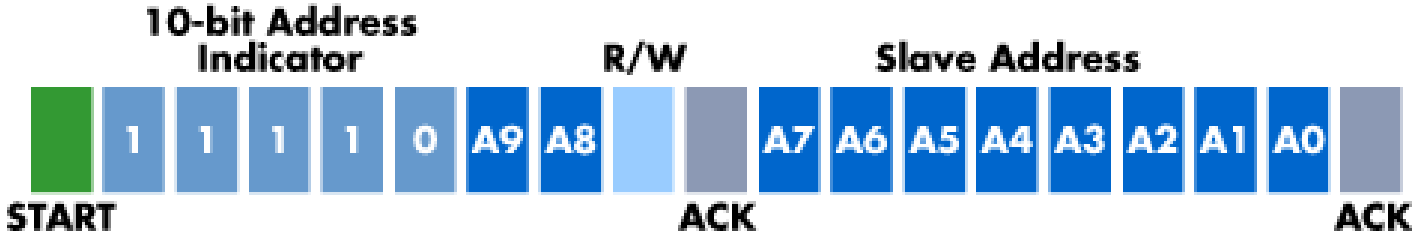
- The 1st byte after START determines the Slave to be addressed
- Some exceptions to the rule:
  - "General Call" address: all devices are addressed : 0000 000 + R/W = 0
  - 10-bit slave addressing : 1111 0XX + R/W = X

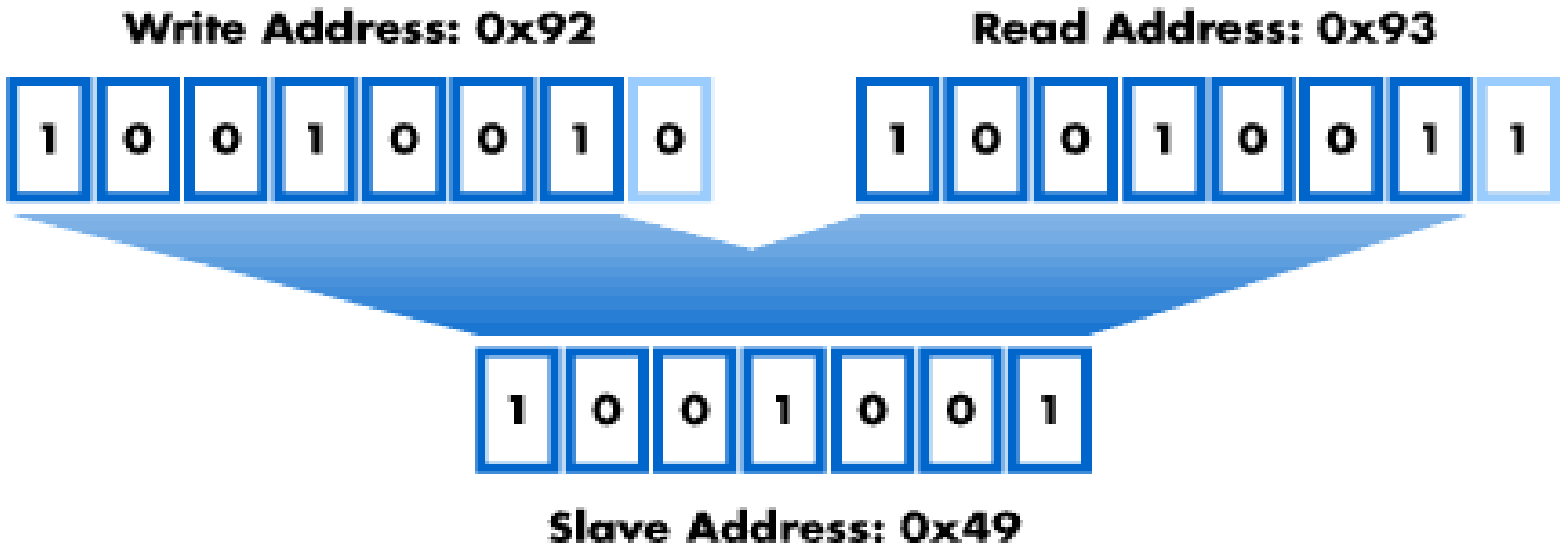
•7-bit addressing



• 10-bit addressing







Reserved Addresses

000 0XXX

0x00 to 0x07

Valid Address Range

0x08 to 0x77

Reserved Addresses

111 1XXX

0x78 to 0x7f